Pancreaticoduodenal Groove: Spectrum of Disease and Imaging Features

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The pancreaticoduodenal groove (PDG) is a small space between the pancreatic head and duodenum where vital interactions between multiple organs and physiologic processes take place. Muscles, nerves, and hormones perform a coordinated dance, allowing bile and pancreatic enzymes to aid in digestion and absorption of critical nutrition. Given the multitude of organs and cells working together, a variety of benign and malignant entities can arise in or adjacent to this space. Management of lesions in this region is also complex and can involve observation, endoscopic resection, or challenging surgeries such as the Whipple procedure. The radiologist plays an important role in evaluation of abnormalities involving the PDG. While CT is usually the first-line examination for evaluation of this complex region, MRI offers complementary information. Although features of abnormalities involving the PDG can often overlap, understanding the characteristic imaging and pathologic features generally allows categorization of disease entities based on the suspected organ of origin and the presence of ancillary features. The goal of the authors is to provide radiologists with a conceptual approach to entities implicating the PDG to increase the accuracy of diagnosis and assist in appropriate management or presurgical planning. They briefly discuss the anatomy of the PDG, followed by a more in-depth presentation of the features of disease categories. A table summarizing the entities that occur in this region by underlying cause and anatomic location is provided.

Abbreviations: CBD = common bile duct, IVC = inferior vena cava, NET = neuroendocrine tumor, PDAC = pancreatic ductal adenocarcinoma, PDG = pancreaticoduodenal groove, PDP = paraduodenal pancreatitis

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SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Describe the various anatomic structures and tissues that interact in the region of the PDG.
- Classify entities occurring in and around the PDG on the basis of their physiopathology.
- Formulate a differential diagnosis for a lesion in the PDG considering its salient characteristics, probable organ of origin, and effect on adjacent structures.

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Introduction

The pancreas, common bile duct (CBD), and duodenum have an intricate relationship along with connective tissue, autonomic nerves, vessels, and lymph nodes. The pancreaticoduodenal groove (PDG) is a thin anatomic space in the superior anterior pararenal space bordered by the pancreatic head medially, duodenal bulb superiorly, descending duodenum laterally, transverse duodenum inferiorly, and inferior vena cava (IVC) posteriorly (Fig 1). Exocrine pancreatic fluid travels in the main pancreatic duct of Wirsung and the accessory duct of Santorini; after joining with biliary secretions from the CBD, it drains through the medial margin of the descending duodenum through the minor and major papillae, respectively. The major papilla is the most distal aspect of the ampulla of Vater, a complex set of sphincters (biliary, pancreatic, and the sphincter of Oddi) that regulate drainage of bile and pancreatic enzymes into the duodenum (Fig 1).

Understanding the numerous cellular lines that compose the structures in and around the PDG is crucial to diagnosis. Three types of mucosal epithelium join and compose the ampulla of Vater: pancreaticobiliary ductal, duodenal intestinal, and foveolar-like mucosa from the ampulla and papillae. Pancreatic endocrine cells produce

TEACHING POINTS

- Sheetlike soft-tissue expansion of the PDG, displacement rather than encasement of vessels and pancreaticobiliary ducts, and small cysts in the PDG or medial duodenal wall are potential indicators of PDP rather than malignancy.
- The differential diagnosis for a solid mass centered in the PDG is similar to that for primary retroperitoneal neoplasms, including lesions of mesodermal, neurogenic, or lymphoid origin.
- Complications of a periampullary duodenal diverticulum include diverticulitis, with inflammatory stranding surrounding the diverticulum; perforation, with extraluminal free air; bezoar formation, with densely packed mottled air and debris; and CBD obstruction, also known as Lemmel syndrome.
- It is important to describe the relationship of PDG lesions to the ampullary complex, involvement of which may necessitate a Whipple procedure.
- When a lesion is located in the PDG itself, expansion of the space between the pancreatic head and duodenum is to be expected, with medial displacement of the pancreatic head and a laterally shifted duodenum.

hormones that circulate in the blood, while acinar exocrine cells and hepatocytes excrete digestive enzymes that travel through the ducts to the duodenum. The duodenal wall is composed of four main layers: mucosa, submucosa, muscularis propria, and serosa or adventitia. Smooth muscle cells and interstitial cells of Cajal exist in the muscularis propria, while Brunner glands and duodenal endocrine cells are found in the submucosa (Fig 2).

Entities Involving PDG

Infectious, Inflammatory, or Malignant Mimics

Inflammation may be centered within the PDG or involve it secondarily owing to its proximity to the pancreas and duodenum.

Paraduodenal pancreatitis (PDP), the preferred term to *groove pancreatitis*, is a form of focal chronic pancreatitis involving the PDG. The pathogenesis of PDP is not entirely clear but is thought to be related to ectopic pancreatic rests in the medial duodenal wall (1). Chronic inflammation in this space results in sheetlike crescentic fibrotic soft-tissue expansion of the PDG, which appears as hypoattenuating and hypovascular at CT—best seen on coronal images—and as T1 hypointense and T2 slightly hyperintense at MRI.

There can be patchy delayed enhancement due to the fibrotic nature of the inflammatory tissue. Other findings include thickening of the medial duodenal wall and cysts in either the PDG or medial duodenal wall, which are best seen at MR cholangiopancreatography (MRCP), although 20%–32% of cases do not have visible cysts (2,3). Two patterns of PDP have been described, depending on whether the inflammatory process is purely confined to the groove or extends to the pancreatic head in a segmental pattern (Fig 3).

Differentiating PDP from pancreatic ductal adenocarcinoma (PDAC) can be challenging but has obvious implications for management and outcomes. The pure form of PDP can mimic PDAC arising in the periampullary region with infiltration into the duodenum. However, the segmental form of PDP can be even more of a diagnostic dilemma for the radiologist. The hypovascular infiltrative nature of PDAC is usually helpful for detection during the arterial or pancreatic phase of CT or MRI, but when healthy pancreatic parenchyma conversely avidly enhances, it can be indistinguishable from chronic forms of pancreatitis, including PDP. Adding to this challenge, one study found that more than half of patients with PDAC have pseudocysts secondary to associated pancreatitis (4), making the presence of cysts alone not pathognomonic for PDP.

A key observation is the effect of the abnormal soft tissue on the pancreatic duct. Smooth narrowing of the pancreatic duct traversing a mass without abrupt or complete obstruction strongly favors an inflammatory process (5). This penetrating duct sign can be best seen at MRCP and is improved by secretin administration. The morphology of the prestenotic pancreatic duct also differs depending on the cause, with contour irregularities and the presence of dilated and deformed side branches present in forms of chronic pancreatitis, as opposed to the severe but smooth upstream dilatation secondary to a malignant stricture.

The well-known double duct sign can be present with either condition; however, abrupt caliber change with shouldering suggests malignancy, while more smooth and tapered narrowing of the ducts with less severe prestenotic dilatation is more commonly due to an inflammatory process. PDAC located within or extending to the PDG will usually encase the pancreaticoduodenal artery and invade adjacent structures (6), while the inflammatory soft tissue in PDP tends to displace vessels (7) (Fig 4). Venous deformation of the confluence, superior mesenteric vein, or portal vein is more likely in PDAC than in PDP, but it is not specific.

Advanced imaging techniques may prove helpful in differentiating the two diagnoses. Studies of perfusion CT and MRI found higher mean blood flow and blood volume in inflammation versus in carcinoma (8). Diffusion-weighted imaging (DWI) studies have shown lower mean apparent diffusion coefficient (ADC) in cases of PDAC

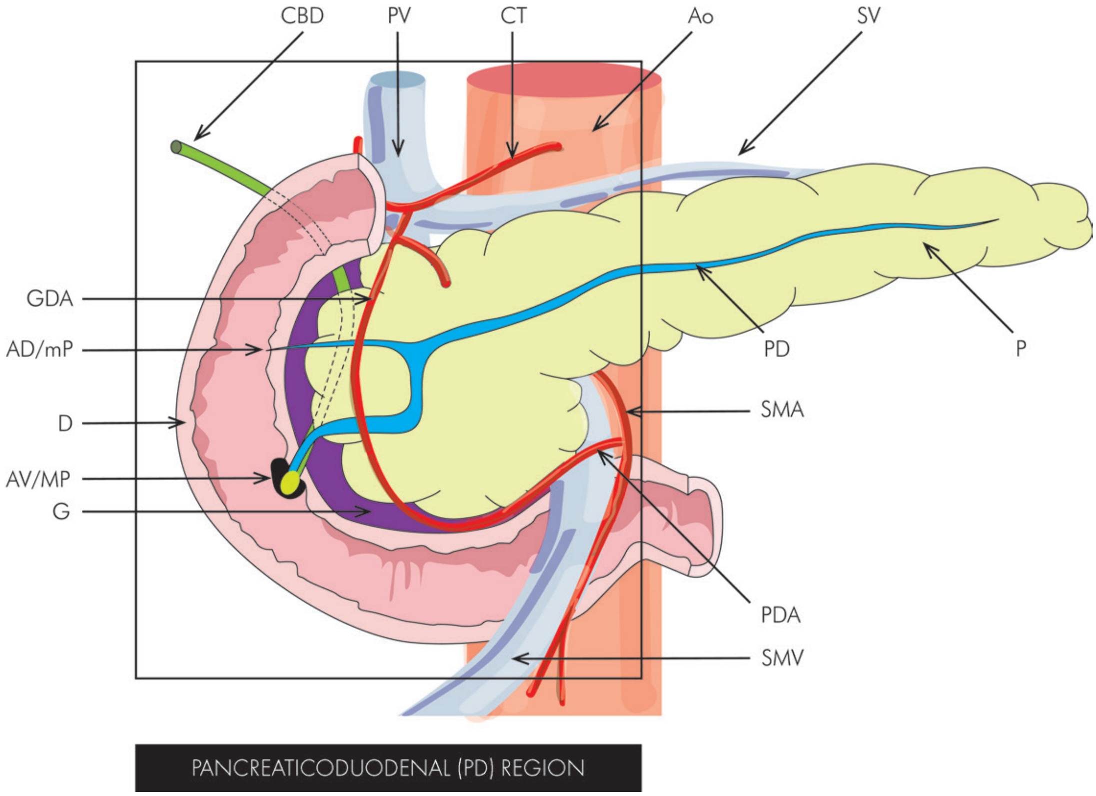


Figure 1. Diagram shows the anatomy of the pancreaticoduodenal region. *AD/mP* = accessory duct and minor papilla, *Ao* = aorta, *AV/MP* = ampulla of Vater and major papilla, *CT* = celiac trunk, *D* = duodenum, *G* = PDG (purple), *GDA* = gastroduodenal artery, *P* = pancreas, *PD* = pancreatic duct, *PDA* = pancreaticoduodenal artery, *PV* = portal vein, *SMA* = superior mesenteric artery, *SMV* = superior mesenteric vein, *SV* = splenic vein.

(9). PDAC has also been found to have a higher standardized uptake value (SUV) at fluorodeoxyglucose (FDG) PET/CT as compared with chronic pancreatic inflammation (10).

Sheetlike soft-tissue expansion of the PDG, displacement rather than encasement of vessels and pancreaticobiliary ducts, and small cysts in the PDG or medial duodenal wall are potential indicators of PDP rather than malignancy. One study found that if three criteria are present—focal thickening of the second portion of the duodenum, cystic changes in the region of the pancreatic accessory duct, and abnormal enhancement of the second portion of the duodenum—distinction of PDP from PDAC can be achieved with high diagnostic accuracy, and the diagnosis of cancer can be excluded with high negative predictive value (11).

Ampullary or duodenal adenocarcinoma may also manifest as infiltrative soft tissue extending to the PDG and be difficult to differentiate from inflammation or PDAC. Ampullary adenocarcinoma can arise from the duodenal epithelium of the ampulla (with behavior similar to duodenal adenocarcinoma), arise from the pancreaticobiliary epithelium of the distal CBD and pancreatic duct (with a worse prognosis), or be a true intra-ampullary tumor with combined duodenal and pancreaticobiliary epithelial morphology, which has the best prognosis of the three, perhaps because of location and early obstruction (12). The three ampullary lineages likely explain the variability in biliary and pancreatic ductal dilatation seen with ampullary adenocarcinoma and cannot be reliably distinguished (13).

Duodenal adenocarcinoma can manifest as a focal mass or involve the duodenal wall circumferentially; may have an infiltrative component involving the adjacent fat planes, encasing vascular structures; and can manifest with nodal or distant metastatic disease. When this tumor arises in close proximity to the ampulla, it can involve and obstruct the biliary and pancreatic ducts and result in the double duct sign, similar to primary ampullary adenocarcinoma (Fig 5).

Other forms of acute or chronic pancreatitis can involve the PDG. Peripancreatic collections can be seen with acute or chronic pancreatitis and include acute peripancreatic collections appearing as ill-defined peripancreatic fluid, which may evolve into encapsulated pseudocysts after about 4 weeks, and acute necrotic collections manifesting as heterogeneous nonliquefied material, which can become encapsulated walled-off necrosis. These collections are usually accompanied by findings of inflammation and lack

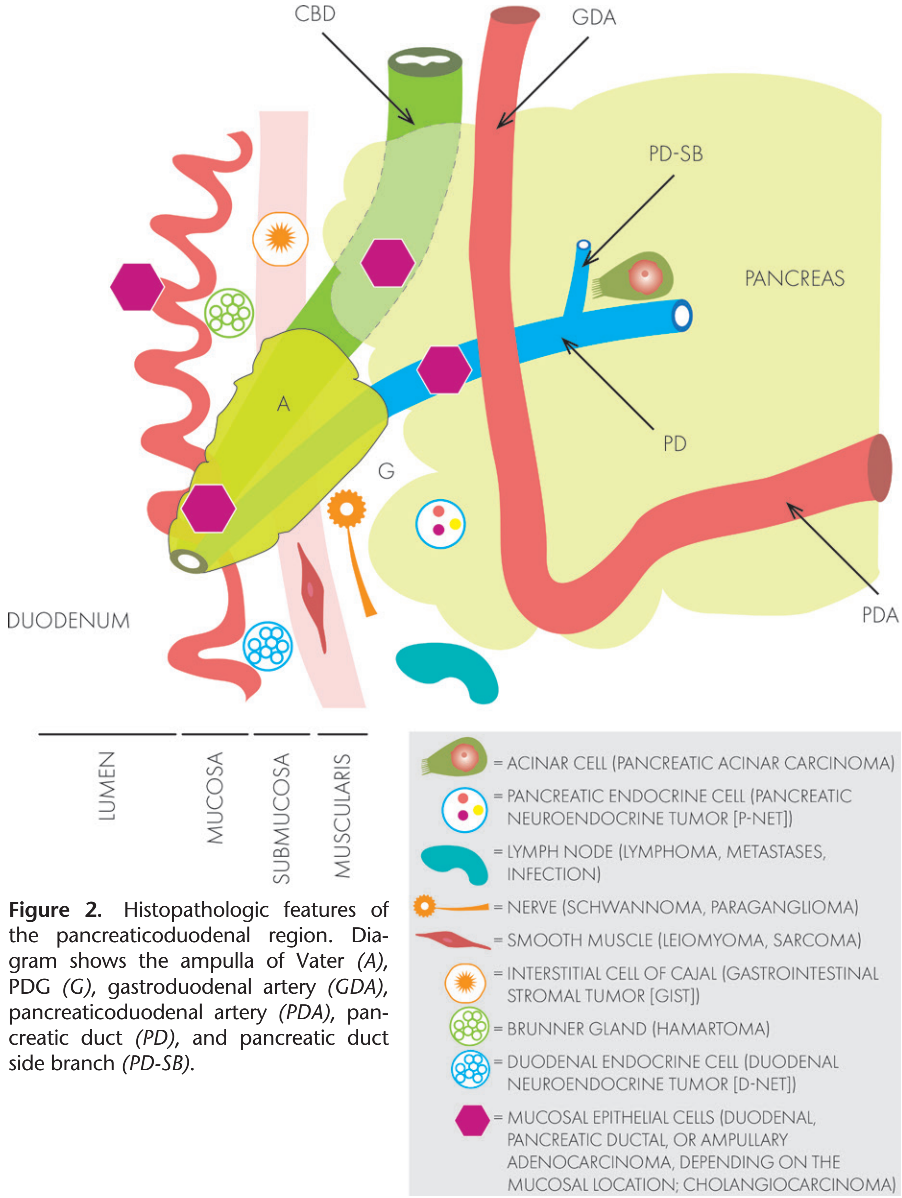


Figure 2. Histopathologic features of the pancreaticoduodenal region. Diagram shows the ampulla of Vater *(A)*, PDG *(G)*, gastroduodenal artery *(GDA)*, pancreaticoduodenal artery *(PDA)*, pancreatic duct *(PD)*, and pancreatic duct side branch *(PD-SB)*.

enhancing components, except for thin linear enhancement of the fibrous capsule in pseudocyst and walled-off necrosis. Although they are most frequently found in the lesser sac, they can occur anywhere in the abdomen and can extend to or be located solely in the PDG (Fig 6).

Duodenal inflammation—evidenced by circumferential wall edema appearing hypoattenuating at CT and hyperintense at T2-weighted imaging, striated hyperenhancing mucosa, and periduodenal stranding—is most commonly seen as reactive to external processes such as acute pancreatitis and cholecystitis. Infectious duodenitis can be due to *Mycobacterium avium-intracellulare* (MAI) or cytomegalovirus (CMV) in the context of HIV infection or AIDS; inflammatory duodenitis can be seen after radiation exposure, with immunoglobulin A (IgA) vasculitis (formerly Henoch-Schönlein purpura), and in 5%–60% of cases of Crohn disease. Findings of duodenitis are nonspecific; therefore, clinical correlation and evaluation of adjacent organs are key to making a unifying diagnosis.

Peptic ulcer disease is usually secondary to infection by *Helicobacter pylori* or prolonged use of nonsteroidal anti-inflammatory drugs (NSAIDs) and most frequently affects the duodenal bulb, which forms the superior border of the PDG. Small ulcers may be occult at imaging, particularly when smaller than 2 cm (14), and most are evident only when leading to perforation, with associated wall thickening, periduodenal fluid, and extraluminal air. Peptic ulcer disease is the most common cause of abnormal gas in the PDG (14), for which radiologists should use multiplanar reconstruction to carefully inspect the duodenal C-loop for focal wall discontinuity or luminal outpouching corresponding to the ulcer crater.

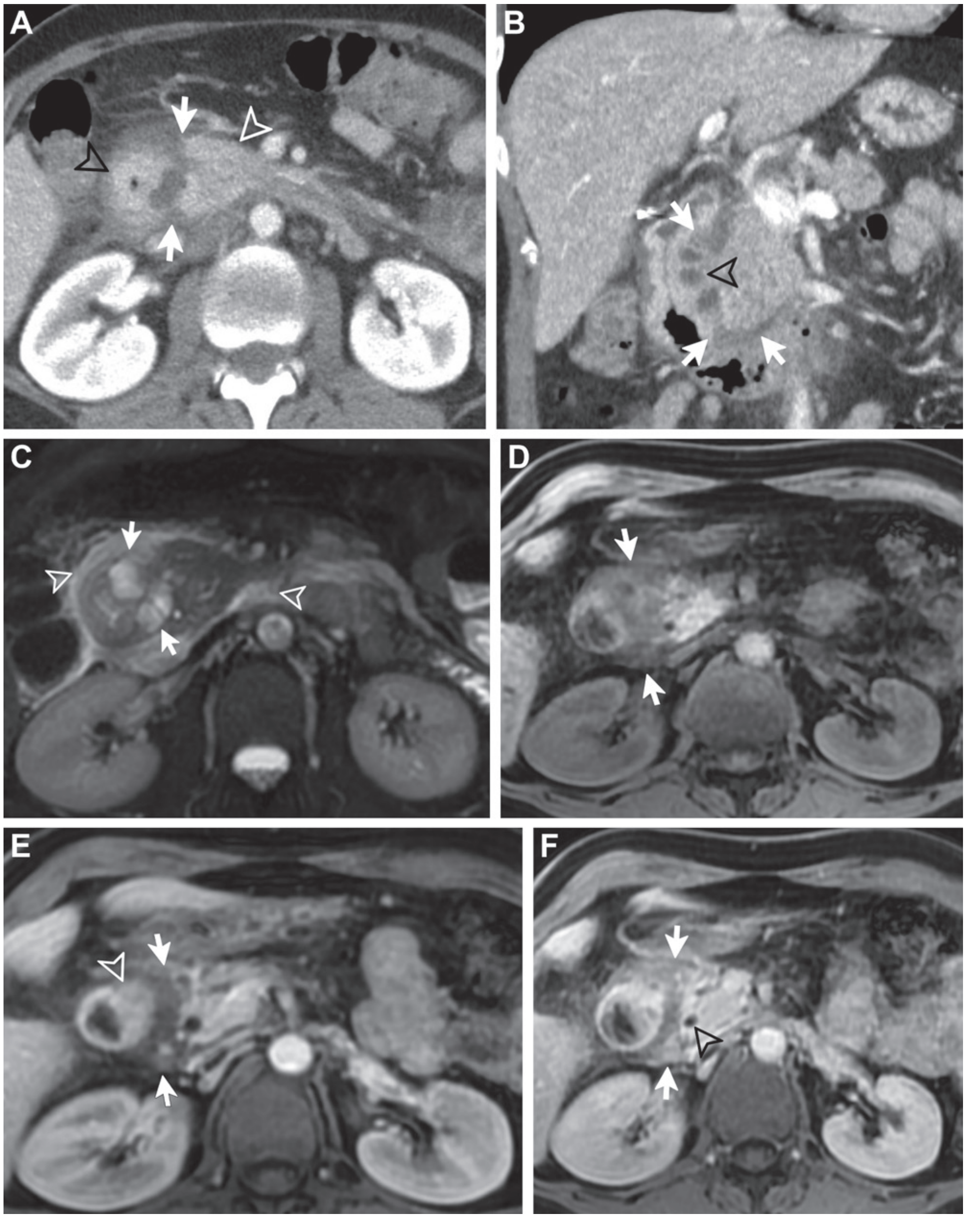


Figure 3. Paraduodenal pancreatitis (PDP) in a 35-year-old man with a history of alcohol use disorder. **(A)** Axial contrast-enhanced CT image shows hypoattenuation in the expanded PDG (arrows), with lateral displacement and wall thickening of the duodenum (black arrowhead) and medial displacement of the pancreatic head (white arrowhead). **(B)** Coronal CT image shows expansion of the PDG (arrows) and rounded cysts along the medial wall of the descending duodenum (arrowhead). Note the caliber narrowing of the duodenal lumen. **(C)** Axial fat-suppressed T2-weighted image shows abnormal hyperintense signal (arrowheads) in a similar distribution as the findings at CT and several cysts (arrows) in the PDG and medial duodenal wall. **(D–F)** Precontrast **(D)**, early postcontrast **(E)**, and late postcontrast **(F)** T1-weighted images show T1-hypointense signal abnormality (arrows) in the PDG with delayed enhancement compatible with fibrosis. Note the asymmetric thickening of the medial duodenal wall (arrowhead in **E**) and the normal-caliber CBD (arrowhead in **F**).

The ulcer may erode into the pancreaticoduodenal artery, causing a pseudoaneurysm, seen as a rounded hyperattenuating or hyperintense structure at CT or MR angiography that follows the attenuation or signal intensity of the blood pool and may result in life-threatening hemorrhage. Peptic ulcer disease may also lead to choledochoduodenal fistula, which should be considered in the setting of abnormal air in the PDG region with pneumobilia. As opposed to gastric ulcers, duodenal ulcers are invariably benign, and endoscopy remains the standard of reference for diagnosis (Fig 7).

Immunoglobulin G4 (IgG4)–related autoimmune disease is a lymphocytic infiltration rich in IgG4-positive plasma cells, resulting in mass-like inflammatory reaction and fibrosis affecting multiple organ systems, including those near the

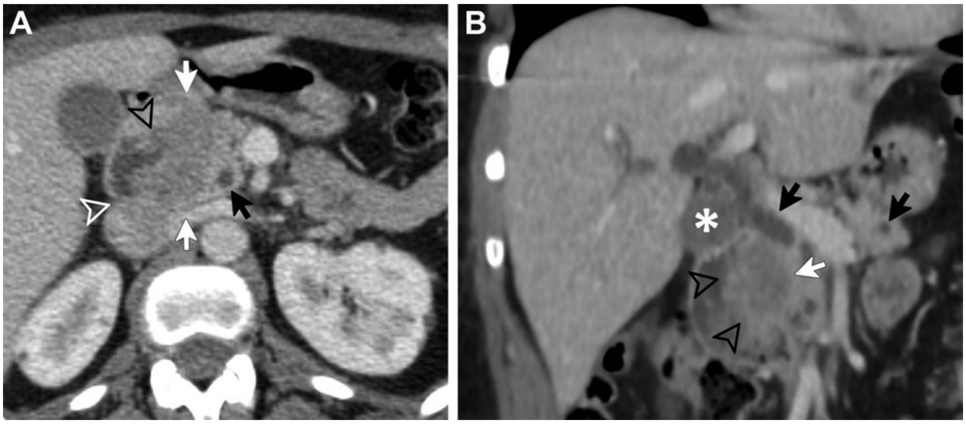


Figure 4. PDAC in a 62-year-old man. **(A)** Axial contrast-enhanced CT image shows a hypoenhancing mass in the PDG (white arrows) with direct invasion of the medial duodenal wall (black arrowhead) and IVC (white arrowhead). Note the normal caliber of the CBD (black arrow). **(B)** Coronal contrast-enhanced CT image shows a hypoenhancing mass in the PDG (white arrow) inseparable from the duodenum, which appears irregular (arrowheads), concerning for direct invasion. Note the metastatic periportal lymphadenopathy (*) and the normal caliber of the CBD and pancreatic duct (black arrows), consistent with sparing of the ducts in this case of medial pancreatic head PDAC.

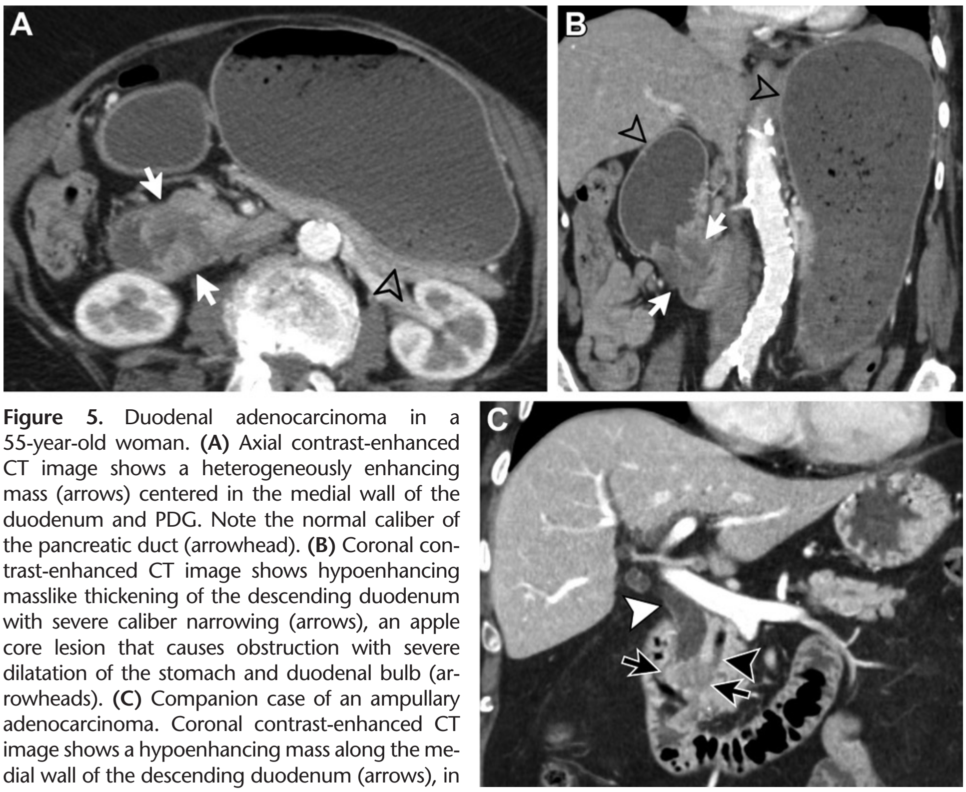


Figure 5. Duodenal adenocarcinoma in a 55-year-old woman. **(A)** Axial contrast-enhanced CT image shows a heterogeneously enhancing mass (arrows) centered in the medial wall of the duodenum and PDG. Note the normal caliber of the pancreatic duct (arrowhead). **(B)** Coronal contrast-enhanced CT image shows hypoenhancing masslike thickening of the descending duodenum with severe caliber narrowing (arrows), an apple core lesion that causes obstruction with severe dilatation of the stomach and duodenal bulb (arrowheads). **(C)** Companion case of an ampullary adenocarcinoma. Coronal contrast-enhanced CT image shows a hypoenhancing mass along the medial wall of the descending duodenum (arrows), in the expected location of the ampulla, with dilatation of the CBD (white arrowhead) and pancreatic duct (black arrowhead), producing the double duct sign.

PDG (15). The pancreas appears enlarged and sausage shaped owing to loss of lobulation and narrowing of the pancreatic duct. Some patients exhibit a peripheral capsulelike rim, which is hypoattenuating at CT and enhances at delayed postcontrast imaging. At MRI, the pancreas demonstrates hypointense T1 signal, beading of the duct, and lack of the true peripancreatic inflammation seen in acute interstitial pancreatitis (16).

Diagnosis is critical, as steroid treatment is effective in both reversing morphologic changes and normalizing pancreatic function (17). Multiorgan involvement can be a clue to IgG4 disease. Other abdominal manifestations include sclerosing cholangitis in up to 88% of cases, retroperitoneal fibrosis in up to 33%, and renal involvement, which can be parenchymal or perirenal (18).

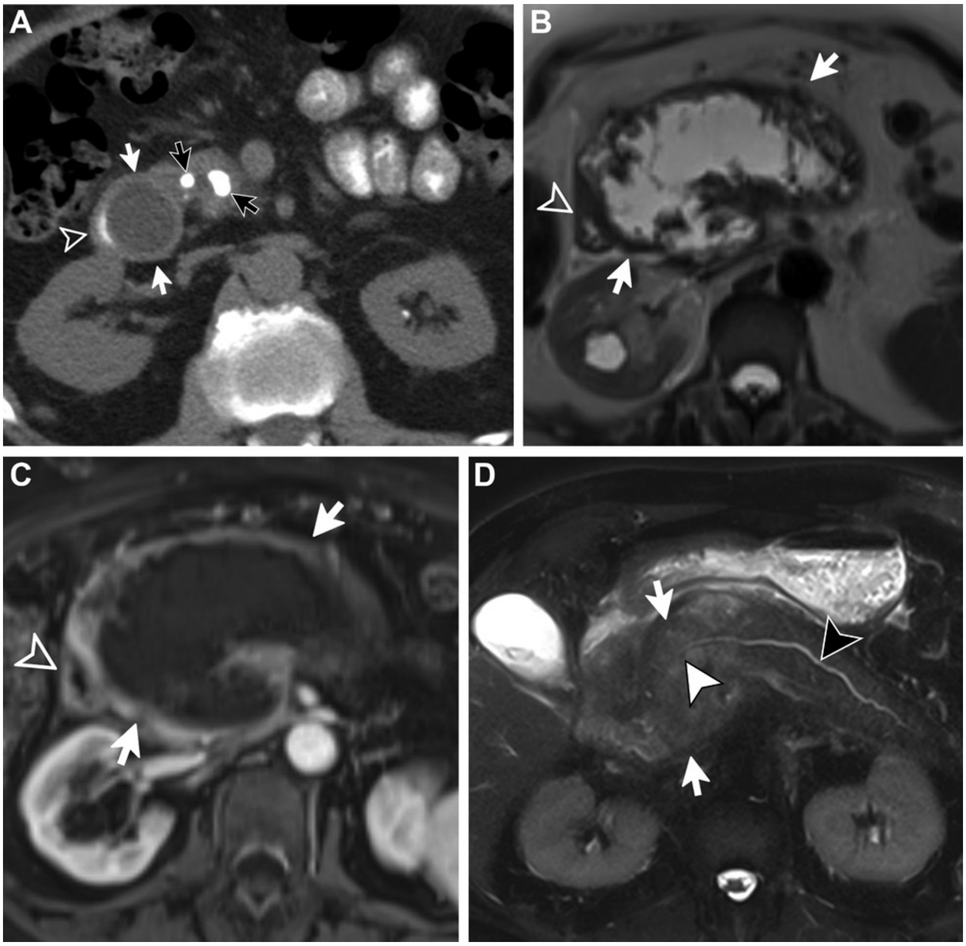


Figure 6. Complications of pancreatitis. **(A)** Pancreatic pseudocyst. Axial noncontrast CT image with oral contrast material shows a rounded hypoattenuating lesion within the medial duodenal wall (white arrows), which effaces the crescent-shaped duodenal lumen containing oral contrast material (arrowhead). Note the pancreatic parenchymal calcifications (black arrows), compatible with chronic pancreatitis. **(B, C)** Walled-off necrosis. Axial T2-weighted image **(B)** shows a collection of high-signal-intensity fluid and debris (arrows), which replaces the normal pancreatic parenchyma. Axial contrast-enhanced T1-weighted image **(C)** shows lack of enhancement except for the thick capsule (arrows). Note the mass effect on the descending duodenum (arrowhead). **(D)** Mass-forming chronic pancreatitis. Axial fat-suppressed T2-weighted image shows masslike expansion of the pancreatic head (arrows), with mild irregularity of the normal-caliber pancreatic duct (black arrowhead) traversing the mass (white arrowhead), producing the penetrating duct sign.

Primary Groove Neoplasms

The differential diagnosis for a solid mass centered in the PDG is similar to that for primary retroperitoneal neoplasms, including lesions of mesodermal, neurogenic, or lymphoid origin, previously described in detail (19–21).

A majority of retroperitoneal neoplasms arise from tissues of mesodermal origin, such as fat (liposarcoma) or smooth muscle (leiomyosarcoma), are often large at presentation, and may involve or compress the PDG (20). Lesion localization and extent of invasion of adjacent structures are crucial for determining the best path for tissue sampling, resectability, and surgical planning. Leiomyosarcoma is often an aggressive tumor, demonstrating partial necrosis, hemorrhage, and invasion of adjacent structures. Most commonly, it is completely extravascular, but it may have an intravascular component when arising from the IVC or renal veins (21) (Fig 8). The presence of calcifications, which can be seen in up to 30% of liposarcomas, is highly uncommon in leiomyosarcoma (20).

The characteristics of liposarcoma vary depending on the degree of histologic dedifferentiation. Well-differentiated tumors show macroscopic fat, which can be easily detected at CT or fat-suppression MRI, have a minimal soft-tissue component, and are well delineated. Dedifferentiated liposarcoma may show complete lack of fat and be impossible to distinguish from leiomyosarcoma.

Desmoid tumor, a form of deep fibromatosis, is a rare mesenchymal tumor that can be sporadic or associated with Gardner syndrome

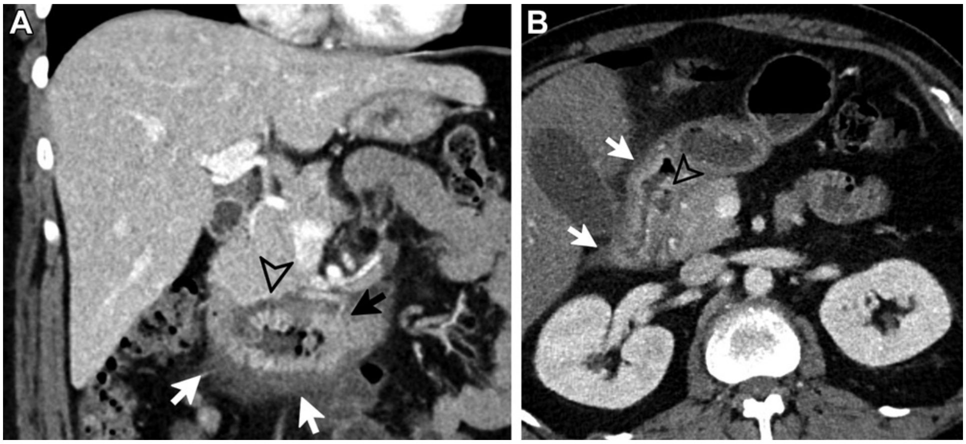


Figure 7. Duodenal inflammation. **(A)** Crohn disease. Coronal contrast-enhanced CT image shows duodenal wall thickening with hyperenhancing mucosa (black arrow), markedly hypoattenuating submucosal edema, and periduodenal stranding (white arrows). Note extension of the edema into the inferior margin of the PDG (arrowhead). **(B)** Peptic ulcer disease. Axial contrast-enhanced CT image shows duodenal wall thickening and periduodenal stranding (arrows). A small duodenal outpouching along the medial wall is seen within the PDG, containing fluid and a tiny focus of gas (arrowhead), consistent with an ulcer.

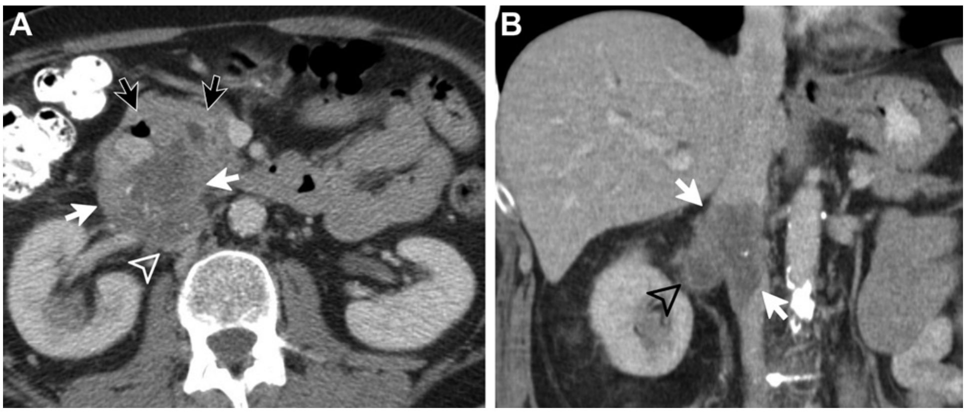


Figure 8. Leiomyosarcoma in a 47-year-old man. **(A)** Axial contrast-enhanced CT image shows an aggressive heterogeneously hypoenhancing mass in the retroperitoneum (white arrows) that involves the IVC (arrowhead), extends into the PDG, and anteriorly displaces the duodenum and pancreatic head (black arrows)—this helps localize the mass as arising posterior to the PDG, rather than within it. **(B)** Coronal contrast-enhanced CT image shows intravascular growth within the IVC (arrows), extending to the right renal vein (arrowhead).

or familial adenomatous polyposis (FAP) and is usually triggered by trauma or surgery (20). Once triggered, well-differentiated spindle cells and collagen proliferate into a locally invasive soft-tissue mass without metastatic potential. This tumor is dependent on estrogen and therefore is more common in females, with peak incidence in the 3rd decade of life (22).

Desmoid tumor appears commonly as a well-circumscribed homogeneously enhancing mass (Fig 9), although it can manifest as infiltrative soft tissue with invasive behavior (23). The appearance depends on the degree of cellular, collagen, and myxoid components at histologic analysis, as well as vascularity. At the early cellular stage, it appears hyperintense at T2-weighted imaging; as the cellularity decreases over time with deposition of collagen, the lesion becomes hypointense at T2-weighted imaging. Contrast-enhanced imaging often reveals diffuse enhancement of variable degree, more pronounced in lesions with increased cellularity.

Treatment options include surgical resection, although the high recurrence rates have made radiation therapy and watchful waiting acceptable alternatives.

Abdominal neurogenic tumors are most commonly located in the retroperitoneum and mostly occur in a younger age group with a better prognosis. Paragangliomas are tumors that arise from chromaffin cells in an extra-adrenal location. The majority are seen in the abdominal para-aortic region, including the Zuckerkandl body, but they can be found in any site with

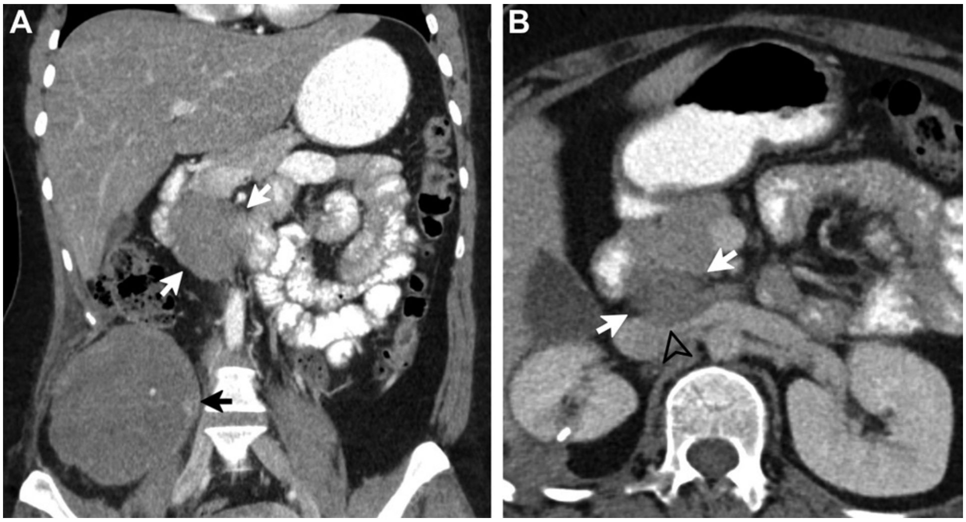


Figure 9. Multiple desmoid tumors in a 52-year-old woman with a history of abdominal trauma. **(A)** Coronal contrast-enhanced CT image shows a homogeneously hypoenhancing mass (white arrows) medial to the descending duodenum and a well-circumscribed larger mass (black arrow) in the right lower quadrant. **(B)** Axial contrast-enhanced CT image shows that the mass (arrows) is located within and posterior to the PDG and has smooth mass effect (arrowhead) on the IVC and left renal vein.

paraganglia, including the PDG. These can be found sporadically or in association with genetic syndromes, including neurofibromatosis type 1 (NF-1), von Hippel–Lindau syndrome (VHL), multiple endocrine neoplasia type 2 (MEN 2), and Carney-Stratakis syndrome.

At CT, paraganglioma is usually seen as a highly vascular well-defined tumor with avid enhancement, punctate calcification, hemorrhage, and areas of cystic degeneration and necrosis (24). Although classically described as "lightbulb bright" on T2-weighted images, this feature is not specific or sensitive and can lead to misdiagnosis in up to 35% of cases (24,25) (Fig 10). Malignancy is common, with 20%–50% having metastases at diagnosis.

Schwannoma is the most common tumor of peripheral nerves and is commonly located in the paravertebral region and, less commonly, adjacent to the PDG, perinephric or presacral space, or abdominal wall. Most cases are solitary and sporadic (90%) and manifest in the 5th or 6th decade. When schwannoma is present in younger patients, an association with neurofibromatosis type 2 (NF-2) is more likely.

At CT, small schwannomas are well defined, homogeneous, and round, but large schwannomas may be heterogeneous with internal degeneration, manifesting as hemorrhage, cystic change, or calcification (Fig 11). At MRI, the appearance depends on the types of tissue they contain: cellular areas show heterogeneous enhancement, and cystic areas are hyperintense on T2-weighted images. Malignant transformation is rare, but progressive enlargement, pain, irregular margins, and infiltration into adjacent tissues are concerning features (26).

Neurofibroma can occur as an isolated tumor or as a component of NF-1. At CT, neurofibroma typically manifests as a well-defined round lesion and appears homogeneously hypoattenuating, owing to the presence of lipid-rich cells. On T1-weighted images, the central portion is hyperintense due to nerve tissue, whereas on T2-weighted images the periphery is hyperintense due to myxoid degeneration.

Pancreaticoduodenal lymphadenopathy is seen as a manifestation of primary lymphoma as well as many metastatic malignancies, although a variety of inflammatory or infectious disorders may involve these nodal chains and mimic malignancy (Fig 12). Lymphoma is the most common primary retroperitoneal malignancy (26), usually manifesting as bulky adenopathy or a confluent soft-tissue mass from coalescing lymph nodes, which may primarily involve the PDG or extend to the space.

At CT, lymphoma manifests as a well-defined homogeneous mass with moderate homogeneous enhancement, which tends to insinuate between and splay or displace normal structures without compressing them. Internal necrosis and gastric outlet or duodenal obstruction can be seen with high-grade tumors. At MRI, lymphoma is usually isointense on T1- and T2-weighted images with moderate homogeneous enhancement. The duodenum and pancreas can rarely be the site of primary lymphoma that grows into the PDG, and imaging-guided biopsy may be required for confirmation of the diagnosis.

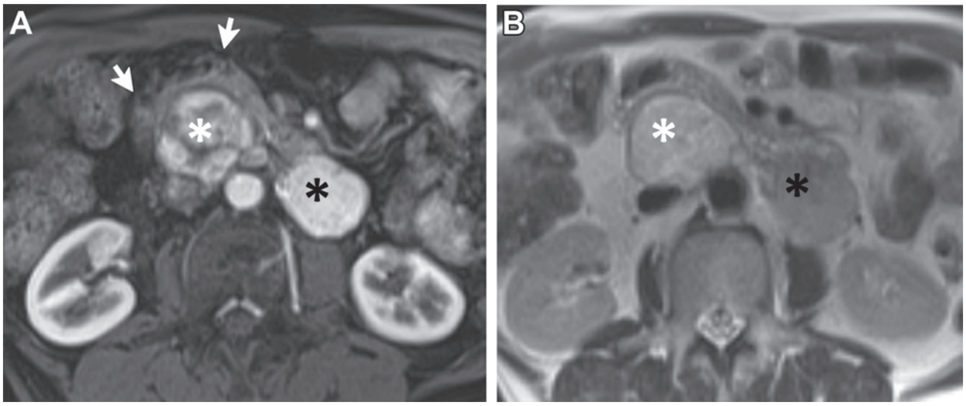


Figure 10. Multiple paragangliomas in a 36-year-old man with a history of multiple endocrine neoplasia type 2 (MEN 2). **(A)** Axial arterial phase contrast-enhanced T1-weighted image shows a well-circumscribed heterogeneously enhancing mass (white ∗) posterior to and anteriorly displacing the pancreas and duodenum (arrows). Note the second, homogeneously enhancing mass in the left retroperitoneum (black ∗). **(B)** Axial T2-weighted image shows hyperintense "lightbulb bright" signal, particularly in the mass (white ∗) located posterior to the PDG. Black ∗ = mass in the left retroperitoneum.

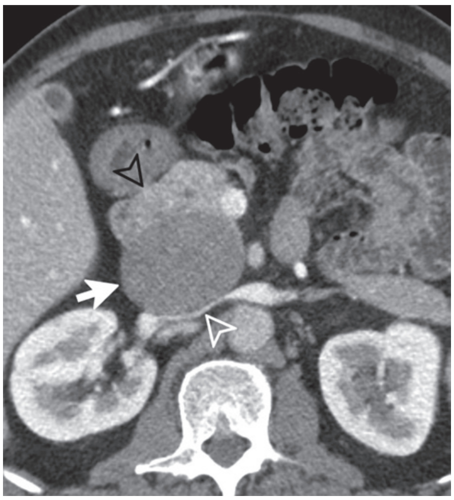


Figure 11. Schwannoma in a 41-year-old man. Axial contrast-enhanced CT image shows a well-circumscribed, homogeneously hypoenhancing, rounded mass (arrow) splaying the duodenum and pancreas anteriorly (black arrowhead) from the IVC posteriorly (white arrowhead), which is compressed.

PDG lymph nodes communicate with gastric, hepatic, and mesenteric nodal chains; therefore, metastatic lymphadenopathy in this region is usually associated with widespread tumor dissemination to the liver and other lymphatic groups. These lymph nodes also receive efferent drainage from the head of the pancreas and duodenum. Isolated hypervascular adenopathy in the PDG may represent metastatic neuroendocrine tumor (NET) and warrants careful evaluation of the gastrinoma triangle, whose vertices are the junction of the second and third portions of the duodenum inferiorly, the intersection of the pancreatic neck and body medially, and the confluence of the cystic duct and CBD superiorly.

Duodenal NETs are rare, often nonfunctional, and incidentally diagnosed at endoscopy. The most common are gastrin cell tumors–gastrinomas, one-third of which are functional and can cause elevated serum gastrin level and Zollinger-Ellison syndrome. Over 50% of gastrinomas are malignant and can metastasize to the regional lymph nodes and other sites. Localization of these tumors is often difficult because of their small size. They may appear as single or multiple arterially enhancing nodules at CT and MRI, more frequently located in the first or second part of the duodenum, with 20% located in the periampullary region. Despite their small size, duodenal NETs can manifest with lymph node metastases in 11%–50% of cases; liver metastases occur late (27) (Fig 13).

Congenital Lesions

Duodenal diverticula are common, seen in approximately 22% of the population in autopsy series, and usually detected incidentally. They may be congenital, or more frequently, acquired pseudodiverticula and periampullary in location, caused by a weakness in the medial duodenal wall from entrance of the CBD or penetrating vessels in this region (28). At CT, these are often seen as rounded air-containing sacs in the pancreatic head or PDG, containing mottled debris or an air-fluid level. At MRI, the air can be detected on in-phase T1-weighted images and gradient-echo T2-weighted images, on which air will create susceptibility.

Complications of a periampullary duodenal diverticulum include diverticulitis, with inflammatory stranding surrounding the diverticulum; perforation, with extraluminal free air; bezoar formation, with densely packed mottled air and debris; and CBD obstruction, also known as Lemmel syndrome (Fig 14). Surgical management

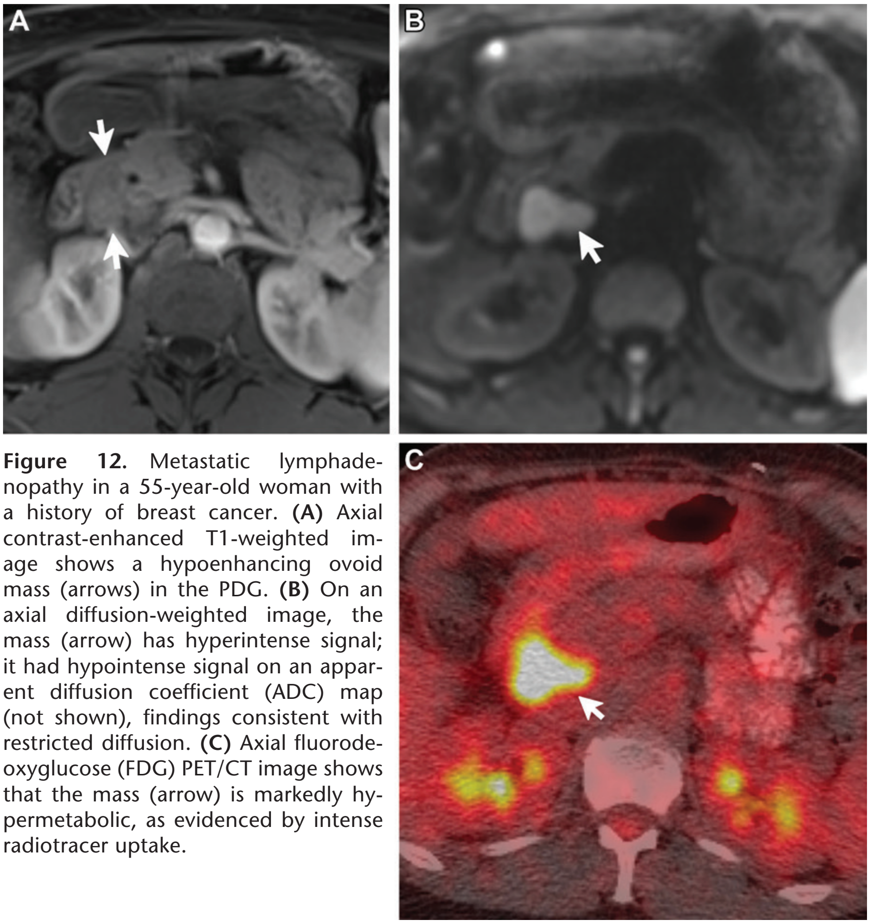


Figure 12. Metastatic lymphadenopathy in a 55-year-old woman with a history of breast cancer. **(A)** Axial contrast-enhanced T1-weighted image shows a hypoenhancing ovoid mass (arrows) in the PDG. **(B)** On an axial diffusion-weighted image, the mass (arrow) has hyperintense signal; it had hypointense signal on an apparent diffusion coefficient (ADC) map (not shown), findings consistent with restricted diffusion. **(C)** Axial fluorodeoxyglucose (FDG) PET/CT image shows that the mass (arrow) is markedly hypermetabolic, as evidenced by intense radiotracer uptake.

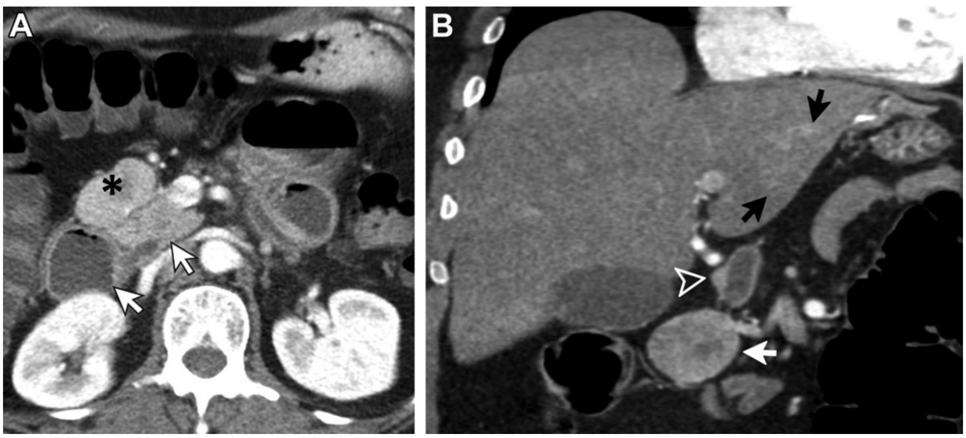
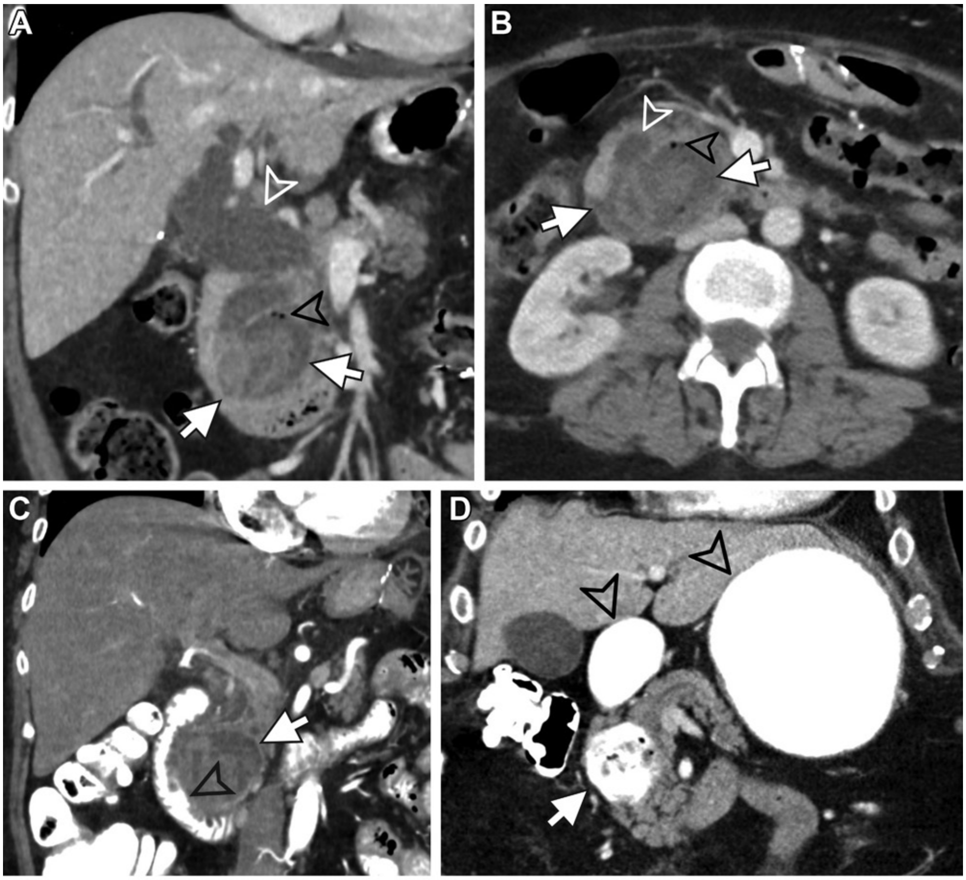


Figure 13. Metastatic duodenal NET in a 49-year-old man. **(A)** Axial arterial phase contrast-enhanced CT image shows an ovoid enhancing mass (*) anterior to the PDG, with slight posterior displacement of the pancreas and duodenum (arrows). **(B)** Coronal CT image shows a small enhancing nodule (arrowhead) along the lateral margin of the duodenal bulb—proved to be a primary duodenal NET—and subtle hepatic lesions (black arrows). The enhancing mass (white arrow) is consistent with a metastatic lymph node.

can be challenging owing to their proximity to the ampulla. Good outcomes have recently been reported with nonoperative management in patients in stable condition with disease contained to the retroperitoneum (28).

Unlike diverticula, enteric duplication cysts are rare true cystic lesions that do not communicate with the duodenal lumen, but may arise from the second and third portions of the duodenum and extend into the PDG. They are smooth thick-walled cysts that share a muscle wall and blood supply with the bowel. They demonstrate simple fluid attenuation at CT and hyperintensity at T2-weighted imaging with a thickened wall. The



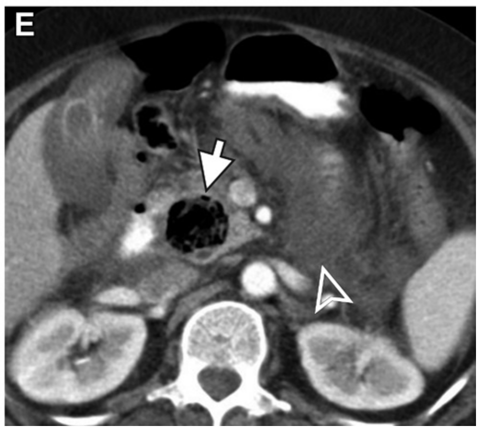


Figure 14. Periampullary duodenal diverticula. **(A–C)** Lemmel syndrome. Coronal **(A)** and axial **(B)** contrast-enhanced CT images show an ovoid heterogeneous periampullary "mass" (arrows) in the PDG that obstructs the CBD (white arrowhead). The key to the diagnosis is the presence of trace gas (black arrowhead) in the diverticulum. Coronal contrast-enhanced CT image with oral and intravenous contrast material **(C)** shows a defect (arrowhead) along the medial descending duodenal wall at the origin of the diverticulum (arrow). **(D)** Gastric outlet obstruction. Coronal contrast-enhanced CT image with oral and intravenous contrast material shows a rounded contrast material–filled periampullary outpouching (arrow) in the PDG, which results in dilatation of the stomach and duodenal bulb (arrowheads). **(E)** Bezoar and pancreatitis. Axial contrast-enhanced CT image with oral and intravenous contrast material shows a rounded lesion (arrow) in the PDG that contains mottled material and air. Note the extensive inflammatory stranding and fluid (arrowhead) in the lesser sac, consistent with acute pancreatitis.

cyst can be centered within the duodenal wall or bulge into the duodenal lumen (Fig 15). Potential complications include biliary obstruction, pancreatitis, torsion, and intussusception.

Type III choledochal cysts represent dilatation of the distal CBD only within the duodenum (also called a choledochocele) and may extend into the PDG, making it sometimes impossible to differentiate them from enteric duplication cysts (29). It is important to distinguish congenital cystic lesions from biliary obstruction due to a mass or pseudocyst; the key to confirming a choledochocele is establishing communication with the biliary tree, which can be done with hepatobiliary phase MRI (Fig 16). Choledochal cysts can also cause pancreatitis or rupture with resultant bile peritonitis and carry a variable risk of malignant transformation; therefore, most are resected.

Annular pancreas is a congenital anomaly that can mimic duodenal wall thickening. It results from failure of migration of the ventral pancreatic bud, resulting in complete or incomplete

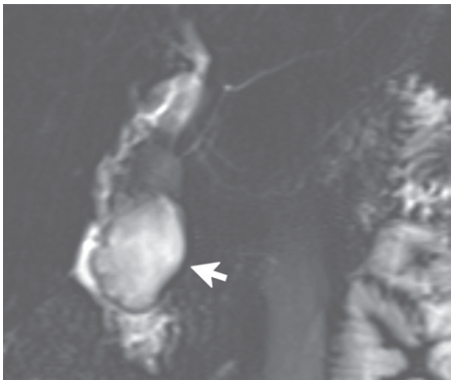


Figure 15. Duodenal duplication cyst in a 22-year-old man. Coronal image from MR cholangiopancreatography (MRCP) shows a lobulated thick-walled cystic mass (arrow) with heterogeneous high signal intensity. The mass protrudes into the duodenal lumen without direct communication.

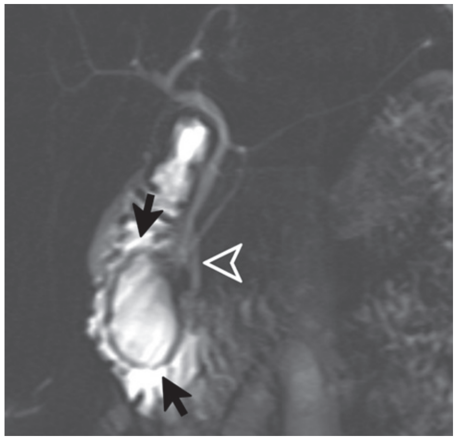


Figure 16. Type III choledochal cyst or choledochocele in a 25-year-old woman. Coronal image from MR cholangiopancreatography (MRCP) shows an intraluminal cyst (arrows) in the descending duodenum. The cyst originates at the ampulla (arrowhead) from the intramural portion of the CBD. The biliary tree and pancreatic duct are otherwise normal.

encasement of the descending duodenum by pancreatic tissue, effectively effacing the PDG. The key to diagnosis is identifying a ring of normal pancreatic tissue—best seen on noncontrast T1-weighted images—where the hyperintense pancreatic tissue wraps around the duodenum (Fig 17).

T2-weighted imaging can also nicely show an aberrant pancreatic duct encircling the duodenum, multiple small ducts entering the duodenum, or pancreas divisum—in fact, there are six subtypes of pancreatic duct branching patterns described in annular pancreas (30). This entity can manifest in infancy with vomiting secondary to duodenal obstruction. In adults, about two-thirds of cases are asymptomatic and found incidentally. When symptomatic, patients can have vague abdominal symptoms and even acute pancreatitis related to the aberrant nature of the ducts (31).

Vascular Lesions

Lymphatic malformation (previously called lymphangioma) is a benign congenital lesion that can be found throughout the body, including the mesentery and less commonly the retroperitoneum. It is usually an asymptomatic, incidental, well-defined, multiloculated macro- or microcystic lesion filled with simple fluid, although it sometimes bleeds internally. At CT, it demonstrates simple fluid attenuation, with a thin lobulated border and thin septa. T2-weighted imaging findings can clinch the diagnosis, as this lesion is very hyperintense owing to its lymph content; it can also be transspatial (Fig 18). Other vascular malformations including low-flow venous malformation and high-flow arteriovenous malformation are infrequently found in the retroperitoneum, where they may rarely involve the PDG.

Aneurysm or pseudoaneurysm of the gastroduodenal artery or superior or inferior pancreaticoduodenal artery can manifest in the PDG, as these vessels pass in and around the groove (Fig 19A–19C). Rupture can lead to fatal outcome in over 40% of cases (32), hence identification is crucial. Pseudoaneurysms can rupture directly into the duodenum, resulting in massive gastrointestinal bleeding with melena, especially when they originate as a complication of peptic ulcer disease. More frequently, they are a complication of pancreatitis and when ruptured cause retroperitoneal or intraperitoneal hematoma.

While catheter angiography is the standard of reference for diagnosis and treatment, CT and MRA angiography allow detection of blood products, as well as a pseudoaneurysm as a small outpouching from an artery. Given the collateral pathways between the celiac artery and superior mesenteric artery (SMA), vascular reconstruction is usually not necessary after surgical ligation unless there is a proximal stenosis. With endovascular embolization, collateral pathways should be considered, given the possibility of preserved collateral flow.

Cavernous transformation of the portal vein occurs with portal vein thrombosis without recanalization, resulting in recruitment of peribiliary venous collaterals, which enlarge and become tortuous. These collaterals can extend to the pancreatic head or PDG and compress the CBD (portal biliopathy) (Fig 19D).

Traumatic Lesions

Abnormal extraluminal air in the PDG after trauma should raise suspicion for traumatic

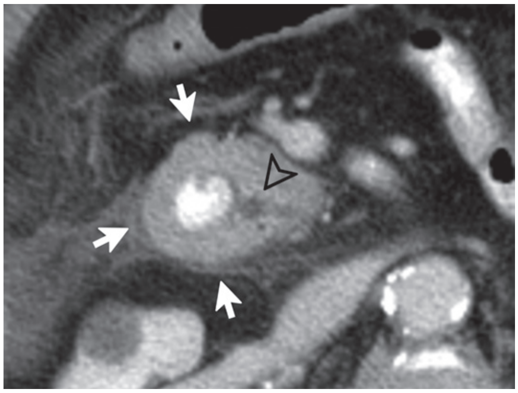


Figure 17. Annular pancreas in a 58-year-old man. Axial contrast-enhanced CT image with oral and intravenous contrast material shows thick enhancing soft tissue (arrows) encasing the second portion of the duodenum, which contains oral contrast material, mimicking duodenal wall thickening. Note the intrapancreatic portion of the CBD (arrowhead) coursing medial to the duodenum.

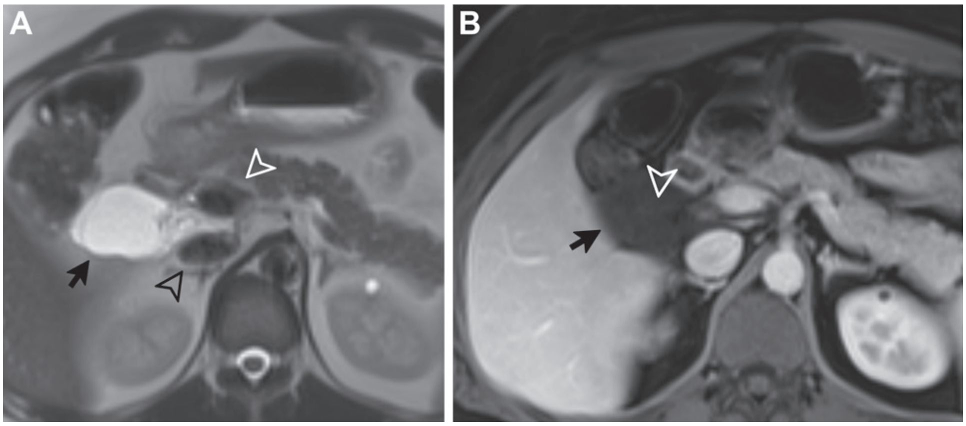


Figure 18. Lymphatic malformation in a 34-year-old man. **(A)** Axial T2-weighted image shows a lobulated high-signal-intensity mass (arrow) in the anterior pararenal space with internal septa and smaller cystic spaces, interposed between the IVC (black arrowhead) and descending duodenum (white arrowhead). **(B)** Axial contrast-enhanced T1-weighted image shows complete lack of enhancement in the lesion (arrow), except for almost imperceptible enhancement of the thin septa (arrowhead).

perforation of the duodenum, which is a surgical emergency and may be the result of penetrating or blunt trauma (33) (Fig 20A). Iatrogenic duodenal perforation is a rare complication of endoscopy and usually treated conservatively, with surgical management reserved for patients with persistent pain, signs of infection, or large perforations of the lateral duodenal wall (34,35).

Posttraumatic or postoperative fluid collections can involve the PDG. Extrahepatic biloma can occur after trauma or intervention, most commonly cholecystectomy. MRI and nuclear medicine agents with biliary excretion can help confirm biliary leak and biloma. Injury to the pancreatic duct with subsequent leak is most frequently a complication of necrotizing pancreatitis (disconnected pancreatic duct syndrome) but can also occur after trauma, where ampullary involvement should be described. MRI with MR cholangiopancreatography (MRCP) after intravenous administration of secretin boosts pancreatic secretions and may help confirm a pancreatic duct leak. Otherwise, endoscopic retrograde cholangiopancreatography (ERCP) with direct visualization is the standard of reference.

Retroperitoneal hemorrhage can be secondary to trauma or aneurysm or pseudoaneurysm rupture or occur spontaneously in patients undergoing anticoagulation therapy. Hemorrhage is seen as hyperattenuating at CT and hyperintense at T1-weighted imaging (Fig 20B, 20C). CT angiography can be used to detect active hemorrhage with extravasation of contrast material. Unless there is active hemorrhage or other complication, hematomas are generally managed conservatively

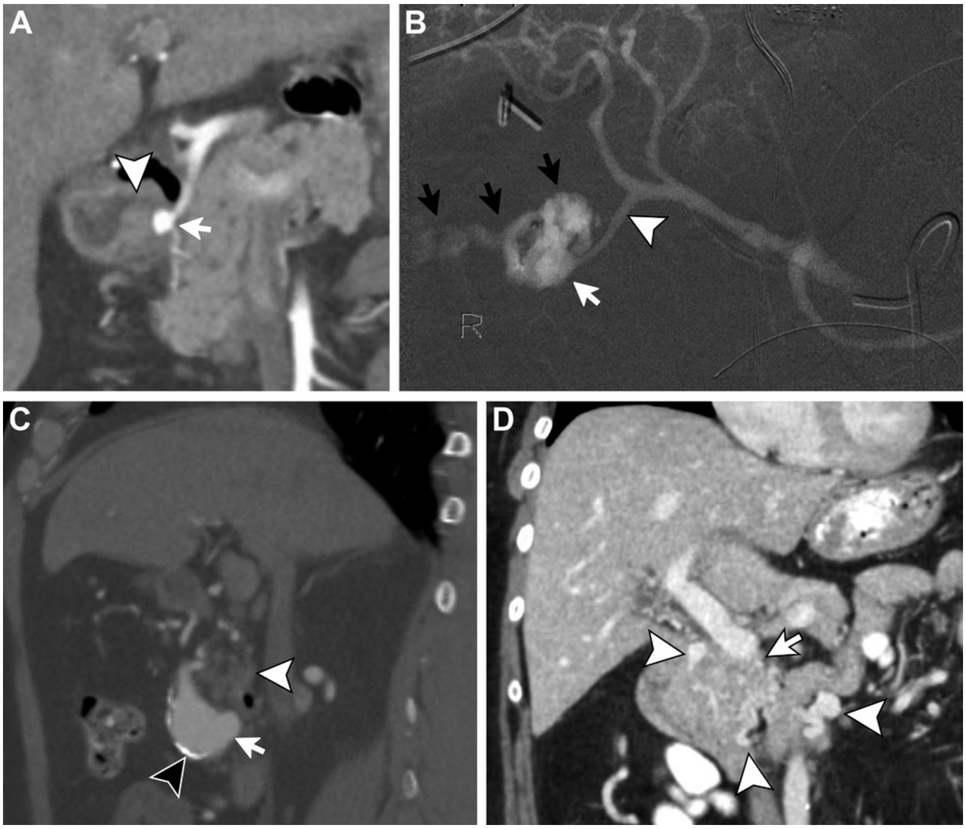


Figure 19. Various vascular entities. **(A, B)** Pseudoaneurysm secondary to peptic ulcer disease. **(A)** Coronal CT angiogram shows a rounded hyperenhancing mass (arrow) near the PDG. A deep penetrating ulcer (arrowhead) can be seen abutting the mass along the medial wall of the proximal duodenum, which is thickened. **(B)** Catheter angiogram shows the pseudoaneurysm (white arrow) arising from the gastroduodenal artery (arrowhead), with active extravasation of contrast material (black arrows) into the ulcer crater and the lumen of the duodenum. **(C)** True aneurysm. Sagittal CT angiogram shows aneurysmal dilatation of the gastroduodenal artery (arrow) near the PDG anterior to the pancreatic head (white arrowhead). True aneurysms more commonly have peripheral calcifications (black arrowhead) owing to their chronicity. **(D)** Cavernous transformation of the portal vein. Coronal contrast-enhanced CT image shows multiple serpentine and tubular enhancing structures (arrowheads) within and surrounding the pancreatic head; these are venous collaterals secondary to chronic occlusion of the portal-mesenteric venous junction (arrow).

with postpyloric tube placement for nutrition; involution can be expected in 1–3 weeks.

Entities Arising from Neighboring Organs That May Extend into PDG

Pancreas

When large enough, lesions in the pancreatic head or uncinate process can extend into the PDG. It is important to know the characteristics of cystic lesions, such as side-branch intraductal papillary mucinous neoplasm (IPMN), mucinous cystic neoplasm (MCN), serous cystadenoma (SCA), and lymphoepithelial cyst (LEC), and of predominantly solid lesions, such as pancreatic NET, solid pseudopapillary epithelial neoplasm (SPEN), and rarer entities including metastases, acinar cell carcinoma, and pancreatic lymphoma. A detailed discussion of these entities is beyond the scope of this article.

Colloid carcinoma of the pancreas is a rare subtype of PDAC (described earlier) characterized by accumulation of extracellular mucin, almost always associated with invasive intestinal-type IPMN (36). It is most frequently located in the pancreatic head, where it can extend into the PDG (37). The mucin appears hypoattenuating at CT and hyperintense at T2-weighted imaging (Fig 21), making it important to differentiate it from true cystic tumors of the pancreas such as IPMN or MCN. A diagnostic clue is meshlike progressive delayed enhancement of a focal hypoenhancing mass with lobulated contours and ill-defined margins (38,39). The 5-year survival rate after resection has been reported as better than that of classic PDAC (40).

Duodenum

The duodenum is an uncommon site of gastrointestinal stromal tumor (GIST), which is a

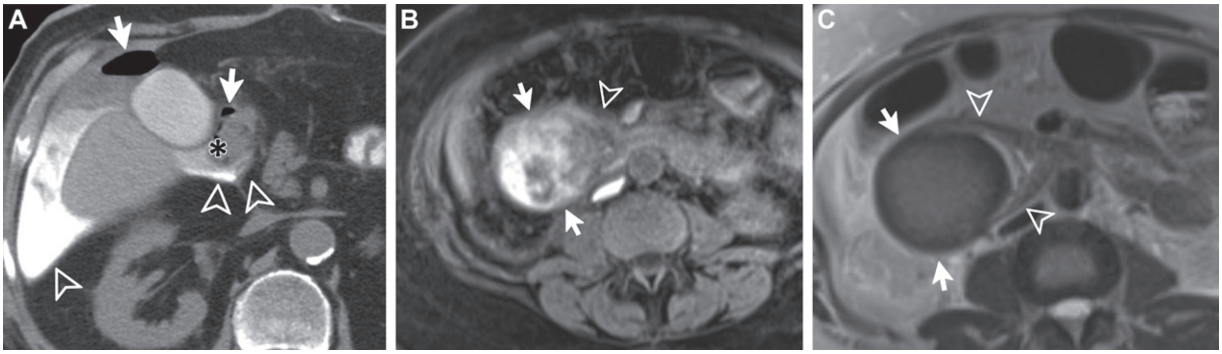


Figure 20. Trauma. **(A)** Duodenal rupture. Axial noncontrast CT image with oral contrast material shows extravasation of oral contrast material (arrowheads) within the PDG and accumulating right laterally in the peritoneal cavity, with dependent gas (arrows) adjacent to the descending duodenum (*) and also in the peritoneal cavity. **(B, C)** Duodenal hematoma. **(B)** Axial noncontrast T1-weighted image shows hyperintense signal (arrows) in the expected location of the descending duodenum and PDG, with medial displacement of the pancreatic head (arrowhead). **(C)** Axial T2-weighted image shows an ovoid mass with hypointense signal centered in the lateral wall of the descending duodenum, as evidenced by the claw sign (arrowheads), with a rim of even lower T2 signal intensity (arrows), narrowing the crescent-shaped duodenal lumen.

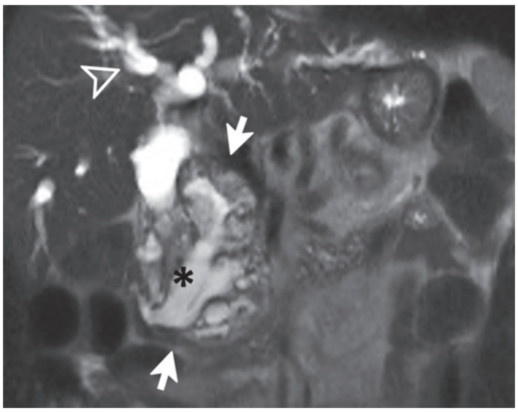


Figure 21. Colloid carcinoma of the pancreas in a 74-year-old man. Coronal T2-weighted image shows a heterogeneously hyperintense mass (arrows) with cystic and solid components centered in the pancreatic head and causing biliary obstruction (arrowhead). Note the marked dilatation of the pancreatic duct (*). At endoscopy, a presumed main-type IPMN was seen bulging into the duodenal lumen and excreting copious mucin (not shown).

Figure 22. GIST. Axial arterial phase contrast-enhanced CT image shows a peripherally enhancing mass in the PDG. The mass arises from the posteromedial descending duodenum with a larger exophytic component (arrow) and smaller endophytic component (arrowhead).

mesenchymal tumor of the gastrointestinal tract arising from the pacemaker interstitial cells of Cajal in the outer third of the muscularis propria. This tumor can be variable in appearance and may have a dumbbell growth pattern with endoluminal and exophytic components, which could potentially mimic a solid mass in the PDG (Fig 22). At CT and MRI, small GISTs typically manifest as a round or lobulated mass with homogeneous moderate enhancement; however, central necrosis and hemorrhage and mucosal ulceration are common features of large GISTs. Patients with neurofibromatosis type 1 (NF-1) have an increased rate of developing duodenal GIST and NET.

Rare malignant mass-forming lesions of the duodenum such as leiomyosarcoma and duodenal lymphoma may infiltrate the PDG, while benign mass lesions such as leiomyoma, adenomatous polyp, and Brunner gland hamartoma may appear to involve it due to their large size. It is important to describe the relationship of PDG lesions to the ampullary complex, involvement of which may necessitate a Whipple procedure.

Common Bile Duct

Most extrahepatic cholangiocarcinomas occur in the proximal third of the duct (perihilar or Klatskin tumors); however, up to 20% originate distal to the cystic duct insertion and tend to manifest with biliary obstruction (41). Periductal infiltrating tumors grow along the duct walls, causing asymmetric wall thickening and enhancement with potential infiltration of the adjacent soft tissues. This results in long-segment caliber narrowing and upstream biliary duct dilatation.

Approach to Entities Involving PDG

Adding to the challenging assessment of this small and complex anatomic space is the fact that conditions arising from it may exhibit overlapping imaging features. It is important for the radiologist to describe the characteristics of the abnormality as well as its relationship to and effect on the adjacent structures.

Identifying the organ of origin or epicenter of a lesion in this space can be problematic. When a lesion is located in the PDG itself, expansion of the space between the pancreatic head and duodenum is to be expected, with medial displacement of the pancreatic head and a laterally shifted duodenum, such as in paraduodenal or groove pancreatitis. On the other hand, if both are displaced in the same direction, the lesion is most likely located outside the groove, such as with a large IVC leiomyosarcoma causing anterior displacement of both structures.

Many times, the epicenter cannot be reliably determined and the abnormality may appear to involve several structures, especially later in its course. In these cases of a large, diffuse, or infiltrative process, clues to the organ of origin may be found in the form of ancillary features. Documenting the presence or absence of biliary and pancreatic duct obstruction and the degree and morphology of upstream ductal dilatation, as well as the effect on adjacent vascular structures including the mesenteric and portal veins and the pancreaticoduodenal artery, can help narrow the differential diagnosis or even favor one entity over another. Rarer ancillary features such as abnormal air, hemorrhage, and fistulization can also be helpful if present.

Conclusion

A variety of inflammatory, structural, or neoplastic entities can affect the PDG (Table). Incorporating a systematic approach based on the nature and location of the abnormality, its effect on adjacent structures, and the presence of ancillary features, while taking into account the histologic diversity of this region, allows radiologists to narrow the differential diagnosis. Providing an accurate interpretation is critical, since it helps direct the best next step in management, which can range from conservative therapy to biopsy and surgical intervention.

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Entities Involving the Pancreaticoduodenal Region by Cause and Organ of Origin

Organ of Origin	Infectious or Inflammatory	Neoplastic	Congenital	Vascular	Traumatic	Obstructive
PDG	PDP IgG4-related disease Lymphadenopathy (including Castleman disease)	Liposarcoma Leiomyosarcoma Desmoid Schwannoma, neurofibroma Paraganglioma Lymphoma Metastatic lymphadenopathy	NA	Vascular malformations (venolymphatic or AVM) GDA or PDA pseudoaneurysm Collateral vessels (including PV cavernous transformation)	Retroperitoneal hemorrhage or hematoma	NA
Pancreas	Acute pancreatitis (including associated collections) Chronic pancreatitis (including mass forming) Autoimmune pancreatitis (IgG4-related disease)	Adenocarcinoma Pancreatic NET (insulinoma, glucagonoma, gastrinoma, VIPoma, nonfunctioning) Acinar cell carcinoma Lymphoma Lipoma Cystic neoplasms (IPMN, SCA, MCN, SPEN) Lymphoepithelial cyst Metastases	Annular pancreas Heterotopic pancreatic tissue Pancreas divisum Santorinicele	NA	Hematoma Laceration or transection Pancreatic duct leak	Annular pancreas Lithiasis
Duodenum	Infectious duodenitis (eg, giardiasis, tropical sprue, *Helicobacter pylori*) Peptic ulcer disease Diverticulitis Crohn disease Celiac disease	Adenocarcinoma Adenoma GIST Duodenal NET (gastrinoma, carcinoid) Leiomyoma or leiomyosarcoma Lymphoma Hamartomatous polyp (ie, Peutz-Jeghers syndrome) Brunner gland hamartoma Hemangioma Lipoma	Diverticulum Enteric duplication cyst	Angiodysplasia Vasculitides: SLE, polyarteritis nodosa, Wegener, IgA (Henoch-Schönlein)	Hematoma Laceration-perforation	Lemmel syndrome Bezoar Tuberculosis
Biliary tree	Complications of cholecystitis (eg, biloma) Choledochoduodenal fistula	Extrahepatic cholangiocarcinoma IPNB Polyp	Choledochal cyst	Portal biliopathy	Bile leak or biloma	Bouveret syndrome Lithiasis
Ampulla-papilla complex	Ampullitis	Adenocarcinoma Adenoma NET (somatostatinoma, ampullary paraganglioma)	NA	NA	NA	Sphincter dysfunction (opiate abuse) Stenosis (eg, HIV)

Note.—AVM = arteriovenous malformation, GDA = gastroduodenal artery, GIST = gastrointestinal stromal tumor, IgA = immunoglobulin A, IgG4 = immunoglobulin G4, IPMN = intraductal papillary mucinous neoplasm, IPNB = intraductal papillary neoplasm of the bile duct, MCN = mucinous cystic neoplasm, NA = not applicable, PDA = pancreaticoduodenal artery, PV = portal vein, SCA = serous cystadenoma, SLE = systemic lupus erythematosus, SPEN = solid pseudopapillary epithelial neoplasm.

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